Closure Requirements for a Compton-Core Vortex-Filament Scaling Model: What Is Fixed Algebraically and What Requires New Dynamics

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(Dated: June 2026)

Abstract

We analyze which parts of a Compton-core vortex-filament model of the electron are fixed algebraically and which parts require additional physical closure. The core postulate identifies the angular rotation rate of the core with the electron Compton frequency, $\Omega_{\text{core}} = \omega_C = m_e c^2/\hbar$. This postulate alone only gives $\|\mathbf{v}_{\circlearrowleft}\| = \omega_C r_c$; it does not determine the characteristic circulation speed, the fine-structure constant, the line tension, or the core density. To close the model one must add: (i) a spectroscopic speed closure equivalent to using the Rydberg constant or α as input; (ii) a core-diameter energy normalization for the line tension; and (iii) a thin-wall Laplace-pressure closure for the density. With these assumptions the scales $\|\mathbf{v}_{\circlearrowleft}\| = (\pi\hbar c R_\infty/m_e)^{1/2}$, $r_c = \|\mathbf{v}_{\circlearrowleft}\|/\omega_C = r_e/2$, $\Gamma_0 = 2\pi r_c\|\mathbf{v}_{\circlearrowleft}\|$, $\mathcal{T}_c = m_e c^2/(2r_c)$, and $\rho_{\text{core}} = m_e c^2/(2\pi\|\mathbf{v}_{\circlearrowleft}\|^2 r_c^3)$ follow consistently. We therefore do not claim a first-principles derivation of α or of ρ_{core}. Instead we identify the minimal missing ingredients required for such a derivation and give phenomenological test templates in terms of a short-range potential correction or an electron form factor.

As a controlled extension, we also state and analyze an explicit spherical pressure-cell closure in which a trefoil Biot–Savart coefficient with conjectured slender-core asymptote $A_K \to 1/(4\pi)$ fixes the local core radius while a spherical volume-curvature response supplies a candidate finite-core correction to the fine-structure scale. The dominant numerical input in this extension is the zeroth-order aspect scale $\mathcal{E}_0 = 274.074996$, which fixes the first four to five significant figures of $\alpha_{\text{cell}} \simeq 2/\mathcal{E}_0$. Thus, within the present paper, the spherical-cell extension is a conditional reproduction given a conjectured external aspect eigenvalue, not a derivation from the Compton-core algebra alone. Companion work replaces this external aspect input by a BEM/NLS-shell finite-cell eigenvalue and treats the far-field coupling as a separate theorem target.

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I. INTRODUCTION

Localized vortex-filament models of particle-like excitations trace their lineage to Helmholtz and Kelvin [1, 2] and continue to appear in topological soliton theory [6], analogue-gravity models [7], and superfluid-vacuum analogies [8]. Such models require at least a core radius, a characteristic circulation speed, a circulation, a line tension, and an effective core density. The central issue is not whether these quantities can be written in terms of known constants; they can. The issue is which relations are independent physical postulates and which are merely algebraic rewritings of standard spectroscopic data.

The present paper therefore adopts a deliberately conservative aim. We ask what follows from a Compton-core hypothesis, what additionally must be supplied to close the model, and what would be required to turn the closure into a first-principles derivation. The result is a hierarchy of assumptions rather than a claimed derivation of the fine-structure constant. In particular, the relation between the circulation speed and α is treated as a spectroscopic speed closure equivalent to using R_∞ as input. Likewise, the line tension and the density require separate mechanical closure assumptions.

This distinction is essential. The Compton-core postulate gives only $\|\mathbf{v}_{\circlearrowleft}\| = \omega_C r_c$. It does not by itself fix either $\|\mathbf{v}_{\circlearrowleft}\|$ or r_c. A genuine derivation of the speed would require an independent dynamical calculation of the electromagnetic coupling, or equivalently of α, without importing R_∞. A genuine derivation of the line tension would require a specified core-energy functional. A genuine derivation of the density would require a specified interface model or stress tensor. These requirements are made explicit below. For reproducibility and referee auditability, Appendix A gives the complete assumption ledger; Appendix B proves the Biot–Savart pressure-balance theorem used for local core closure; Appendix C derives the spherical pressure-cell Jacobian used in the closure factor; Appendix D states the resulting conditional fine-structure closure theorem; Appendix E records the numerical convergence and robustness requirements; Appendix F gives the sensitivity and falsification checks; and Appendix G quarantines historical numerical coincidences that are not used as evidence in the derivation. A companion spherical/NLS pressure-cell paper derives the pressure scale E_p^{NLS} and shell stiffness $\kappa_{\mathrm{NLS}}^{\mathrm{shell}}$, while a companion BEM/NLS finite-cell paper numerically extracts the stationary aspect eigenvalue $E_\star$. Those results are not assumed in the algebraic closure

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analysis below; they address precisely the new dynamics identified here as missing. The accompanying supplementary script `reproduce_alpha_cell_closure.py` implements the minimal numerical audit pipeline used in this paper: Fourier ideal-trefoil loading, Biot–Savart cutoff-energy scanning, extraction of A_K, the no-contact check $a_{\rm nc}/r_c = \sqrt{4\pi A_K}$, and the final spherical-cell evaluation of $\alpha_{\rm cell}$; see Appendix E for the reproducibility protocol.

We also record a more ambitious closure candidate motivated by the ideal-trefoil robustness tests. In that candidate the local Biot–Savart logarithmic coefficient is assigned the conjectured slender-core asymptote $1/(4\pi)$, giving the core-radius closure $a = r_c$, while the remaining small correction to a zeroth-order topological aspect ratio is attributed to the spherical pressure response of the surrounding irrotational cell. The aspect scale $\mathcal{E}_0 = 274.074996$ is not derived in this paper; it is a conjectured external aspect eigenvalue whose independent derivation is an open problem. Since $\alpha_{\rm cell} \simeq 2/\mathcal{E}_0$ at leading order, this input fixes the first four to five significant figures of the final number. The spherical-cell construction should therefore be read as a conditional closure/reproduction, not as a first-principles derivation from a complete microscopic action.

Throughout we use SI units, with $\hbar = h/(2\pi)$, m_e, c, and e carrying their CODATA 2022 recommended meanings [3]. CODATA 2022 is based on a least-squares adjustment of data available through 31 December 2022 [3]. We denote the circulatory velocity field by $\mathbf{v}_{\circlearrowleft}$ and its magnitude by $\|\mathbf{v}_{\circlearrowleft}\| = \|\mathbf{v}_{\circlearrowleft}\|$.

II. THE COMPTON-CORE POSTULATE

Postulate 1 (Compton-core hypothesis)**.** *The vortex core undergoes solid-body rotation at the electron Compton angular frequency,*

$$\Omega_{\rm core} = \omega_C \equiv \frac{m_e c^2}{\hbar}. \tag{1}$$

The magnitude of the circulatory velocity at the core boundary of radius r_c is therefore

$$\|\mathbf{v}_{\circlearrowleft}\| = \omega_C \, r_c. \tag{2}$$

Remark 1 (Physical motivation)**.** *For a solid-body rotating core of radius r_c and angular velocity Ω, the tangential speed at the boundary is $\|\mathbf{v}_{\circlearrowleft}\| = \Omega \, r_c$. The Compton frequency $\omega_C = m_e c^2/\hbar$ is the natural frequency associated with the electron rest energy: it is the* *frequency at which one quantum $\hbar\omega_C = m_e c^2$ is stored. Postulate 1 is the simplest resonance condition consistent with the electron rest energy as the core-energy scale, and it provides a geometric link between the rotation rate and the core boundary. It is a hypothesis about the core dynamics, not a consequence of standard quantum electrodynamics, and it is testable: any independent measurement of r_c inconsistent with $r_c = \|\mathbf{v}_\circlearrowleft\|/\omega_C$ (where $\|\mathbf{v}_\circlearrowleft\|$ is determined by Sec. V) falsifies the postulate.*

III. THE SPECTROSCOPIC SPEED CLOSURE

A. Equivalence of R_∞ and α as inputs

The Rydberg constant satisfies

$$R_\infty = \frac{\alpha^2 m_e c}{2h} = \frac{\alpha^2 m_e c}{4\pi\hbar}. \tag{3}$$

Thus R_∞ and α carry the same information once $\{m_e, c, h\}$ are specified. Using R_∞ instead of α is therefore not an independent derivation of α; it is a convenient spectroscopic parametrization of the same dimensionless input.

Closure 1 (Spectroscopic speed closure)**.** *The characteristic circulation speed is related to the electromagnetic coupling by*

$$\tilde{\alpha} \equiv \frac{2\|\mathbf{v}_\circlearrowleft\|}{c}, \qquad R_\infty = \frac{\tilde{\alpha}^2 m_e c}{2h}. \tag{4}$$

Equivalently,

$$R_\infty = \frac{\|\mathbf{v}_\circlearrowleft\|^2 m_e}{\pi\hbar c} = \frac{\|\mathbf{v}_\circlearrowleft\|^2 \omega_C}{\pi c^3}. \tag{5}$$

Remark 2 (What would be needed to derive the speed)**.** *Closure 1 is the strongest non-geometric input in the model. A first-principles derivation would require a dynamical calculation of the dimensionless coupling*

$$\tilde{\alpha} = \mathcal{F}[\mathcal{I}_1, \mathcal{I}_2, \ldots], \tag{6}$$

where the invariants $\mathcal{I}_i$ are defined by an explicit field-theoretic or hydrodynamic action. Unless Eq. (6) can be evaluated without using R_∞, atomic spectra, or the CODATA value of α, the speed remains calibrated rather than derived.

IV. CLOSURE OF THE CHARACTERISTIC SPEED

Using Closure 1,

$$R_\infty = \frac{\|\mathbf{v}_\circlearrowleft\|^2\,\omega_C}{\pi c^3} \quad\Longrightarrow\quad \|\mathbf{v}_\circlearrowleft\|^2 = \frac{\pi\,\hbar\,c\,R_\infty}{m_e}. \tag{7}$$

Proposition 1 (Characteristic circulation speed)**.** *Given Closure 1 and the measured R_∞,*

$$\boxed{\|\mathbf{v}_\circlearrowleft\| = \sqrt{\frac{\pi\,\hbar\,c\,R_\infty}{m_e}}} \tag{8}$$

is the unique positive closure value for the tangential circulation speed. The inputs are $\{m_e, c, \hbar, R_\infty\}$; since $R_\infty \equiv \alpha^2 m_e c/(2h)$, this is equivalent to using α as input. The result is therefore calibrated by spectroscopy, not predicted independently.

Dimensional check

$$\left[\sqrt{\hbar\,c\,R_\infty/m_e}\right] = \sqrt{\frac{\text{J·s}\cdot\text{m·s}^{-1}\cdot\text{m}^{-1}}{\text{kg}}} = \sqrt{\text{m}^2\,\text{s}^{-2}} = \text{m}\,\text{s}^{-1}.\checkmark$$

Numerical value

$\|\mathbf{v}_\circlearrowleft\| = 1.093\,845\,631\,5\times 10^{6}\,\text{m}\,\text{s}^{-1}$ (from CODATA 2022 [3]).

V. DERIVED MECHANICAL SCALES

The following scales follow algebraically from Proposition 1 and Postulate 1, each with a direct classical-hydrodynamic interpretation.

A. Core radius and its relation to the classical electron radius

$$r_c = \frac{\|\mathbf{v}_\circlearrowleft\|}{\omega_C} = \frac{\|\mathbf{v}_\circlearrowleft\|\,\hbar}{m_e c^2} = 1.408\,970\,16\times 10^{-15}\,\text{m}. \tag{9}$$

Remark 3 (Relation to the classical electron radius)**.** *The classical electron radius is*

$$r_e = \frac{e^2}{4\pi\varepsilon_0 m_e c^2} = \frac{\alpha\hbar}{m_e c} = 2.817\,940\,3\times 10^{-15}\,\text{m}. \tag{10}$$

From (9) *and* $\alpha = 2\|\mathbf{v}_\circlearrowleft\|/c$ *(established below),* $r_c = \alpha\hbar/(2m_e c) = r_e/2$*. We emphasize that* r_c *is an* effective model parameter*, not a claim about the physical extent of the electron. The classical radius* r_e *itself plays an analogous role in QED — appearing in Thomson cross-sections, Compton scattering, and vacuum polarization — without implying a physical "size." Experimental upper bounds on the electron's spatial extent (from* $g-2$ *and high-energy scattering measurements,* $r \lesssim 1 \times 10^{-18}\,\mathrm{m}$ *[21]) constrain the* point-like *quantum-mechanical extent of the charge distribution, not the scale at which an effective vortex-core model is constructed. For a toroidal idealization in which the tube centre lies at distance* $R = r_c$ *from the symmetry axis and the tube has minor radius* $a = r_c$ *(horn-torus geometry), the outer boundary lies at* $R + a = 2r_c = r_e$*. This gives a geometric interpretation of the factor-of-two relation, but it is not evidence that the electron charge has a resolved radius* r_e*.*

B. Fine-structure constant as input equivalence

Proposition 2 (Algebraic equivalence of α and $\|\mathbf{v}_\circlearrowleft\|$)**.** *The ratio* $\tilde{\alpha} \equiv 2\|\mathbf{v}_\circlearrowleft\|/c$ *satisfies* $\tilde{\alpha} = \alpha$ *(the CODATA fine-structure constant) because Closure 1 was imposed. This is not a derivation of* α *from independent physics but the algebraic consequence of* $R_\infty \equiv \alpha^2 m_e c/(2h)$ *combined with* (8)*.*

Proof. $\tilde{\alpha} = (2/c)(\pi\hbar c R_\infty/m_e)^{1/2}$. Substituting $R_\infty = \alpha^2 m_e c/(4\pi\hbar)$: $\tilde{\alpha} = (2/c)(\alpha^2 c^2/4)^{1/2} = \alpha$. □ □

Numerically, $\tilde{\alpha} = 7.297\,352\,567\,1 \times 10^{-3}$ versus CODATA $\alpha = 7.297\,352\,569\,3 \times 10^{-3}$; the residual 3×10^{-10} is the propagated uncertainty from R_∞ (relative uncertainty 1.9×10^{-12}) combined with the rounding of $\{m_e, c, \hbar\}$. It is a precision-consistency check, not a prediction.

C. Circulation quantum

The line integral of velocity around the core boundary defines the circulation,

$$\Gamma_0 \equiv \oint \mathbf{v}_\circlearrowleft \cdot d\mathbf{l} = 2\pi\, r_c\, \|\mathbf{v}_\circlearrowleft\| = \frac{2\pi\|\mathbf{v}_\circlearrowleft\|^2}{\omega_C} = 9.683\,619 \times 10^{-9}\,\mathrm{m^2\,s^{-1}}. \tag{11}$$

The ratio $\Gamma_0/\kappa = \alpha^2/4 \approx 1.33 \times 10^{-5}$, where $\kappa = h/m_e$ is the Onsager-Feynman quantum [10, 11], showing that the present circulation is smaller by $\alpha^2/4$.

D. Vortex line tension

The *vortex line tension* is the energy stored in the core per unit filament length [9]. It is not fixed by Postulate 1. We therefore introduce a separate normalization closure.

Closure 2 (Core-diameter energy normalization)**.** *One electron rest-energy quantum is assigned to a filament segment of length $2r_c$:*

$$\mathcal{T}_c \equiv \frac{m_e c^2}{2r_c} = \frac{\hbar\omega_C}{2r_c}. \tag{12}$$

The physical content of Closure 2 is that the core stores one quantum of Compton energy $m_e c^2 = \hbar\omega_C$ per core-diameter length. A first-principles derivation would require an explicit energy functional whose on-shell line energy equals Eq. (12).

Corollary 1 (Line tension in fundamental constants)**.**

$$\mathcal{T}_c = \frac{m_e c^2}{2r_c} = \frac{m_e^2 c^3}{\tilde{\alpha}\,\hbar} = 29.053\,51\,\mathrm{N}. \tag{13}$$

Remark 4 (Disambiguation from the gravitational tension scale)**.** *The quantity $F_{\mathrm{gr}} \equiv c^4/(4G)$ defines the maximum force transmissible without forming a causal horizon in general relativity [12, 13]. It and $\mathcal{T}_c$ share the dimension of force (energy per unit length), but are conceptually distinct: $\mathcal{T}_c$ is a mechanical property of the vortex core, while F_{gr} is a causal upper bound from spacetime physics. Numerically $\mathcal{T}_c \approx 29\,\mathrm{N}$ versus $F_{\mathrm{gr}} \approx 3\times 10^{43}\,\mathrm{N}$. Their ratio*

$$\frac{\mathcal{T}_c}{F_{\mathrm{gr}}} = \frac{4\alpha_g}{\tilde{\alpha}} \approx 9.60\times 10^{-43}, \tag{14}$$

where $\alpha_g = G m_e^2/(\hbar c) \approx 1.752\times 10^{-45}$, involves only fundamental constants and has no free parameters.

E. Core mass density from Laplace-pressure balance

The density is also not fixed by the Compton-core postulate. We close it with a thin-wall pressure-balance model.

Closure 3 (Thin-wall Laplace-pressure closure)**.** *For a cylindrical core of radius r_c and line tension $\mathcal{T}_c$, define an effective wall tension*

$$\gamma \equiv \frac{\mathcal{T}_c}{2\pi r_c}. \tag{15}$$

The corresponding cylindrical Laplace pressure is

$$P_L = \frac{\gamma}{r_c} = \frac{\mathcal{T}_c}{2\pi r_c^2}. \tag{16}$$

We then impose the mechanical balance

$$\frac{1}{2}\rho_{\text{core}}\|\mathbf{v}_{\circlearrowleft}\|^2 = \frac{\mathcal{T}_c}{2\pi r_c^2}. \tag{17}$$

Closure 3 is a thin-interface assumption. It is not a generic theorem of vortex dynamics; other core stress tensors or outer cutoffs would change the numerical coefficient.

Corollary 2 (Core density)**.** *Solving* (17) *and substituting* (12)*:*

$$\boxed{\rho_{\text{core}} = \frac{\mathcal{T}_c}{\pi r_c^2\|\mathbf{v}_{\circlearrowleft}\|^2} = \frac{m_e c^2}{2\pi\,\|\mathbf{v}_{\circlearrowleft}\|^2\, r_c^3}.} \tag{18}$$

Numerically, $\rho_{\text{core}} = 3.8934 \times 10^{18}\,\text{kg}\,\text{m}^{-3}$.

Dimensional check

$[m_e c^2/(\|\mathbf{v}_{\circlearrowleft}\|^2 r_c^3)] = \text{kg}\,\text{m}^2\,\text{s}^{-2}\,/\,(\text{m}^2\,\text{s}^{-2}\cdot\text{m}^3) = \text{kg}\,\text{m}^{-3}$. ✓

Remark 5 (Status of the density result)**.** *Equation* (18) *is non-circular only because Closure 2 fixes* $\mathcal{T}_c$ *before Closure 3 is applied. It is nevertheless not a first-principles density derivation. It is the density implied by the chosen tension normalization and the chosen thin-wall pressure law.*

For comparison, the usual local-induction estimate for the line energy of a vortex filament contains a circulation, a density, and typically a logarithmic dependence on an outer length scale [9]. Therefore, any future derivation of ρ_{core} *from a hydrodynamic action must specify the core profile, the outer cutoff or renormalization rule, and the stress tensor. Without those ingredients, Eq.* (18) *should be read as a closure consequence, not as an independent prediction.*

F. Geometric gate identity

Lemma 1 (Geometric gate)**.**

$$\frac{\lambda_c}{\pi r_c} = \frac{4}{\tilde{\alpha}} \approx 548.14, \tag{19}$$

where $\lambda_c = h/(m_e c)$ *is the electron Compton wavelength.*

Proof. $\lambda_c = 2\pi\hbar/(m_e c)$; $r_c = \tilde{\alpha}\hbar/(2m_e c)$: $\lambda_c/(\pi r_c) = 4/\tilde{\alpha}$. □ □

This dimensionless ratio is pure algebra; it is a geometric property of the length scales λ_c and r_c and does not encode additional physical content.

VI. DERIVATION CHAIN AND HYDRODYNAMIC INTERPRETATION

The complete derivation chain is:

$$\underbrace{m_e,\ c,\ \hbar,\ R_\infty}_{\text{standard inputs}} \xrightarrow{\text{Closure 1}} \|\mathbf{v}_\circlearrowleft\| \xrightarrow{\text{Post. 1}} r_c \to \Gamma_0 \xrightarrow{\text{Closure 2}} \mathcal{T}_c \xrightarrow{\text{Closure 3}} \rho_{\text{core}}. \tag{20}$$

The classical-hydrodynamic interpretation of each scale is:

$\|\mathbf{v}_\circlearrowleft\| = \omega_C r_c$	tangential speed at solid-body core boundary
$r_c = \|\mathbf{v}_\circlearrowleft\|/\omega_C$	core radius from solid-body rotation
$\Gamma_0 = 2\pi r_c \|\mathbf{v}_\circlearrowleft\|$	circulation (line integral of velocity)
$\mathcal{T}_c = m_e c^2/(2r_c)$	line tension from core-diameter energy closure
ρ_{core}	core density from thin-wall pressure-balance closure

VII. STATUS TABLE

VIII. WHAT WOULD BE REQUIRED FOR A FIRST-PRINCIPLES DERIVATION

The preceding sections define a consistent closure hierarchy. A full derivation would require replacing each closure by a dynamical equation. The minimal requirements are as follows.

A. Independent derivation of the speed or fine-structure constant

One must specify an action or Hamiltonian containing both the circulatory degrees of freedom and the electromagnetic coupling, for example schematically

$$S = \int d^4x\, \mathcal{L}\big[\mathbf{v}_\circlearrowleft, \rho, A_\mu, \psi; \Pi_i\big], \tag{21}$$

with dimensionless parameters or topological invariants Π_i. The model must then predict

$$\alpha = \mathcal{F}(\Pi_i), \qquad \|\mathbf{v}_\circlearrowleft\| = \frac{\alpha c}{2}, \tag{22}$$

TABLE I. Epistemic status of all quantities. "Closure" denotes an explicit model assumption beyond the Compton-core postulate; "Algebraic" denotes a consequence of earlier rows.

Quantity	Symbol	Status	Note
Compton frequency	$\omega_C = m_e c^2/\hbar$	Standard def.	[3]
Core rotation rate	$\Omega_{\text{core}} = \omega_C$	Postulate	Post. 1
Rydberg/FSC input	$R_\infty \equiv \alpha$	Empirical input	(3)
Speed relation	$2\|\mathbf{v}_\circlearrowleft\|/c = \tilde{\alpha}$	Closure	Closure 1
Circ. speed	$\|\mathbf{v}_\circlearrowleft\|$	Calibrated	Prop. 1
Core radius	$r_c = r_e/2$	Post.+closure	(9)
Circulation	Γ_0	Algebraic	(11)
Line tension	$\mathcal{T}_c$	Closure	Closure 2
Core density	ρ_{core}	Closure result	Closure 3
Geom. gate	$\lambda_c/(\pi r_c)$	Algebraic	Lemma 1
Trefoil log coeff.	A_K	Numerical/asymptotic ansatz	App. E
Aspect scale	$\mathcal{E}_0 = 274.074996$	Conjectural input	App. A
Cell-radius coeff.	$\chi_R = 2$	Closure choice	App. F
Spherical cell factor	Ξ_{sph}	Closure candidate	App. C
Cell reproduction for FSC	α_{cell}	Conditional	App. D
Background density	ρ_f	Open	needs independent coupling
Topological state	knot type K	Open	not determined here

without using R_∞, atomic spectra, or the CODATA value of α. This is the missing step if the goal is to derive the speed rather than calibrate it.

B. Candidate spherical pressure-cell closure for the fine-structure constant

This subsection gives a concrete closure candidate for the missing step in Eq. (22). It is deliberately separated from the algebraic closure chain above. The result is conditional: if the spherical pressure-cell functional below is the correct coarse-grained outer-cell functional, and if the external aspect input $\mathcal{E}_0$ is accepted, then the model reproduces the numerical fine-structure scale. If either ingredient is not derived from the full dynamics, the result remains a conjectural closure. The full assumption ledger, pressure theorem, spherical-cell Jacobian, numerical checks, and sensitivity analysis are given in Appendices A–F.

Let K denote the ideal trefoil. The local irrotational contribution to the Biot–Savart energy of a slender filament is written in the normalized form

$$E_{\mathrm{BS}}(a) = \rho_f \Gamma_0^2 L_K \left[A_K \log\left(\frac{L_K}{a}\right) + a_K \right], \tag{23}$$

where a is the tube radius, L_K is the physical centerline length, and A_K is the logarithmic local-induction coefficient. The normal pressure on a tube wall of area $2\pi a L_K$ is

$$P_{\mathrm{irr}}(a) = -\frac{1}{2\pi a L_K}\frac{\partial E_{\mathrm{BS}}}{\partial a} = \frac{\rho_f \Gamma_0^2 A_K}{2\pi a^2}. \tag{24}$$

Balancing this against the core dynamic pressure

$$P_{\mathrm{core}} = \frac{1}{2}\rho_f \|\mathbf{v}_{\circlearrowleft}\|^2 \tag{25}$$

and using $\Gamma_0 = 2\pi r_c \|\mathbf{v}_{\circlearrowleft}\|$ gives

$$\frac{a}{r_c} = \sqrt{4\pi A_K}. \tag{26}$$

Thus the conjectured slender-core asymptotic value $A_K = 1/(4\pi)$ is exactly the no-contact core-radius closure $a = r_c$. This is also the closure convention used in the numerical robustness script, where $a_{\mathrm{nc}} = r_c\sqrt{4\pi A_K}$ and the stationary roots are solved in $x = a/r_c$ rather than directly at meter scale. Finite-resolution numerical runs need not equal $1/(4\pi)$ exactly; the use of $1/(4\pi)$ below is an asymptotic ansatz, not a fitted finite-resolution value.

The remaining question is whether the dimensionless speed ratio can be predicted. We introduce a zeroth-order topological aspect ratio $\mathcal{E}_0$ and write

$$\alpha_{\mathrm{cell}} = \frac{2}{\mathcal{E}_{\mathrm{eff}}}, \qquad \mathcal{E}_{\mathrm{eff}} = \mathcal{E}_0(1-\delta_{\mathrm{cell}}). \tag{27}$$

At leading order, $\alpha_{\mathrm{cell}} \simeq 2/\mathcal{E}_0$. Consequently the choice of $\mathcal{E}_0$ determines the first four to five significant figures of the final value, while the pressure-cell correction below operates only at relative order $\mathcal{E}_0^{-2} \sim 10^{-5}$. In the present manuscript $\mathcal{E}_0$ is therefore classified as a conjectured external aspect eigenvalue, not as a quantity derived from the pressure-cell calculation itself. A fundamental version of the model must derive $\mathcal{E}_0$ independently, as discussed in Sec. VIII and Appendix A.

The local trefoil calculation supplies the asymptotic factor $\pi^2 A_K \to \pi/4$. The spherical cell supplies the finite-volume correction multiplying this local factor. Let

$$\mathcal{L}_K \equiv \frac{L_K}{D_K} \tag{28}$$

be the dimensionless ropelength of the ideal trefoil, and define the small spherical shell parameter

$$\eta_K \equiv \frac{1}{4\mathcal{L}_K}. \tag{29}$$

More generally one may write the spherical control radius as $R_{\text{cell}} = \chi_R L_K$. Then

$$\eta_K(\chi_R) = \frac{a}{R_{\text{cell}}} = \frac{1}{2\chi_R \mathcal{L}_K}. \tag{30}$$

The numerical closure below uses $\chi_R = 2$, giving $\eta_K = 1/(4\mathcal{L}_K)$. The choice $\chi_R = 2$ is a model closure, not derived in this paper; Appendix F records the associated sensitivity.

We now postulate the minimal dimensionless pressure-cell Gibbs functional

$$\mathcal{G}_{\text{sph}}(q; \eta_K) = \frac{K_{\text{sph}}}{2}\, q^2 - K_{\text{sph}}\left(1 + 3\eta_K + \eta_K^2\right) q, \qquad K_{\text{sph}} > 0, \tag{31}$$

where q is the dimensionless multiplier of the local trefoil correction $\pi^2 A_K/\mathcal{E}_0^2$. This is a virtual-work/enthalpy closure: the quadratic term penalizes departure from the isotropic cell response, and the linear term is the coarse-grained pressure work performed by the shifted spherical control surface. The three forcing terms are assigned as follows: 1 is the local tube-pressure contribution, $3\eta_K$ is the first variation of a spherical volume element $V \propto R^3$, and η_K^2 is the leading quadratic curvature-shell response retained after removing the purely local wall term already accounted for by A_K. The combination is still a closure rule. A more general weighted form is analyzed in Appendix F; only a primitive stress-tensor derivation could promote Eq. (31) to a first-principles result.

Proposition 3 (Spherical pressure-cell closure)**.** *For the functional* (31)*, the unique stationary and stable pressure-cell multiplier is*

$$\Xi_{\text{sph}}(\chi_R) = 1 + 3\eta_K(\chi_R) + \eta_K(\chi_R)^2. \tag{32}$$

For the closure choice $\chi_R = 2$, this becomes

$$\Xi_{\text{sph}} = 1 + 3\eta_K + \eta_K^2 = 1 + \frac{3}{4\mathcal{L}_K} + \frac{1}{16\mathcal{L}_K^2}. \tag{33}$$

Consequently, if the corrected aspect ratio is

$$\mathcal{E}_{\text{eff}} = \mathcal{E}_0\left[1 - \frac{\pi}{4}\frac{\Xi_{\text{sph}}}{\mathcal{E}_0^2} + O(\mathcal{E}_0^{-4}, \mathcal{L}_K^{-3})\right], \tag{34}$$

then the predicted coupling is

$$\boxed{\alpha_{\text{cell}} = \frac{2}{\mathcal{E}_0\left[1 - \frac{\pi}{4\mathcal{E}_0^2}\left(1 + \frac{3}{4\mathcal{L}_K} + \frac{1}{16\mathcal{L}_K^2}\right)\right]}}. \tag{35}$$

Proof. Stationarity gives

$$\frac{\partial \mathcal{G}_{\rm sph}}{\partial q} = K_{\rm sph}\left[q - \left(1 + 3\eta_K + \eta_K^2\right)\right] = 0, \tag{36}$$

so $q = \Xi_{\rm sph}$. Since

$$\frac{\partial^2 \mathcal{G}_{\rm sph}}{\partial q^2} = K_{\rm sph} > 0, \tag{37}$$

the stationary point is a strict minimum. Substituting $A_K = 1/(4\pi)$ into the local finite-core factor $\pi^2 A_K/\mathcal{E}_0^2$ gives $(\pi/4)/\mathcal{E}_0^2$. Multiplication by the stable spherical-cell response $\Xi_{\rm sph}$ yields Eq. (34), and $\alpha_{\rm cell} = 2/\mathcal{E}_{\rm eff}$ gives Eq. (35). □

For numerical orientation, take

$$\mathcal{E}_0 = 274.074996, \qquad \mathcal{L}_K = 16.371498, \qquad \chi_R = 2. \tag{38}$$

Here $\mathcal{E}_0$ is a conjectured external aspect input. The value $\mathcal{L}_K = 16.371498$ is the ropelength of the ideal trefoil with tube diameter $D_K = 1$ used in the numerical ideal-trefoil run and is taken from the Knot Atlas ideal-knot Fourier data file [18]; the rigorous mathematical ropelength framework is reviewed in Ref. [17]. Then

$$\eta_K = 1.52704413 \times 10^{-2}, \qquad \Xi_{\rm sph} = 1.04604451, \tag{39}$$

$$\mathcal{E}_{\rm eff} = 274.071998421, \qquad \alpha_{\rm cell}^{-1} = 137.035999211. \tag{40}$$

Compared with the CODATA 2022 value of α, this numerical value is within 2.4×10^{-10} fractionally. This level of agreement should not be overinterpreted. The dominant zeroth-order scale is supplied by $\mathcal{E}_0$, because $\alpha_{\rm cell} \simeq 2/\mathcal{E}_0$; the spherical-cell factor only supplies the residual 10^{-5} finite-core correction. Thus the extension is a conditional reproduction given $\mathcal{E}_0$, χ_R, and $\mathcal{L}_K$, not yet a derivation from a complete microscopic action. Its falsifiable content is the pair of coefficients $3/(4\mathcal{L}_K)$ and $1/(16\mathcal{L}_K^2)$ together with the radius choice $\chi_R = 2$; changing the pressure-cell geometry changes the prediction.

C. Independent derivation of the line tension

A line tension must follow from an energy functional. For a cylindrical core profile this would require an expression of the form

$$\mathcal{T}_{\rm dyn} = \int_0^{R_{\rm out}} 2\pi r\, dr \left[\frac{1}{2}\rho(r)v_\theta(r)^2 + U(\rho, \nabla\rho) + \mathcal{E}_{\rm int}(r)\right], \tag{41}$$

plus a renormalization prescription for the outer scale $R_{\rm out}$. Closure 2 is derived only if

$$\mathcal{T}_{\rm dyn} = \frac{m_e c^2}{2r_c} \tag{42}$$

follows as the stationary on-shell value.

D. Independent derivation of the density

The density requires a stress-balance calculation. A sufficient route is to define a radial stress tensor σ_{ij} and show that the normal stress jump at the core wall gives

$$\Delta P = \sigma_{rr}^{(\rm out)} - \sigma_{rr}^{(\rm in)} = \frac{\gamma}{r_c}, \qquad \gamma = \frac{\partial E}{\partial A}, \tag{43}$$

with $\gamma = \mathcal{T}_c/(2\pi r_c)$ as a consequence rather than a definition. Only then does $\rho_{\rm core}$ become dynamically predicted.

E. Independent low-energy observable

Finally, the model must produce an observable correction. A generic low-energy parametrization is

$$F_e(q^2) = 1 - \frac{1}{6}\langle r^2\rangle_{\rm eff} q^2 + \mathcal{O}(q^4), \qquad \langle r^2\rangle_{\rm eff} = \chi r_c^2, \tag{44}$$

where χ must be predicted. Equivalently, the theory may predict a short-range potential correction

$$\Delta V_\eta(r) = \eta \, \frac{e^2}{4\pi\varepsilon_0 r} \, f(r/r_c), \qquad f(0) \text{ finite}, \quad f(x \gg 1) \to 0, \tag{45}$$

where η and f must be fixed by the model. Without such a coefficient or shape function, the core scale is not experimentally sharp.

IX. PHENOMENOLOGICAL TESTS AND FALSIFIERS

The present closure model does not yet make a parameter-free prediction for atomic spectra. It does, however, define test templates that become falsifiable once the coefficients in Eqs. (44) and (45) are fixed.

(i) Spectroscopic potential test. For any specified $\Delta V_\eta(r)$, hydrogenic energy shifts are

$$\Delta E_{n\ell} = \int_0^\infty |R_{n\ell}(r)|^2 \, \Delta V_\eta(r) \, r^2 \, dr. \tag{46}$$

Existing bound-state QED calculations already account for the Lamb shift to high precision [14, 15]. Therefore, any order-unity η producing an unobserved shift is excluded. If the model predicts $\eta \ll 1$, the corresponding bound must be stated.

(ii) Form-factor test. If the model predicts χ in Eq. (44), then scattering data constrain the effective charge response. The scale $r_c \simeq 1.409 \times 10^{-15}$ m cannot be interpreted naively as a resolved electron radius; it is only observable through the predicted coupling coefficient χ.

(iii) Tension-ratio diagnostic. The ratio

$$\frac{\mathcal{T}_c}{F_{\mathrm{gr}}} = \frac{4\alpha_g}{\tilde{\alpha}} \tag{47}$$

is a useful dimensionless diagnostic, but not by itself a falsifiable prediction. It becomes testable only after a concrete analogue-system mapping identifies the measured mechanical tension and the analogue causal-horizon scale.

X. NUMERICAL SUMMARY

The numerical values in Table II are accompanied by the standalone supplementary script `reproduce_alpha_cell_closure.py`. The script is not used as a substitute for the assumptions in Appendix A; it is only a reproducibility aid for the Biot–Savart coefficient extraction, the local closure ratio, and the spherical-cell numerical evaluation described in Appendix E.

XI. DERIVED-LABEL POLICY FOR COMPANION RESULTS

The companion pressure-cell and BEM/NLS papers should not be read as changing the epistemic classification inside the present closure audit. A quantity may be labelled first-principles derived only if all coefficients entering it are obtained from a stated variational or boundary-value problem without empirical fine-structure input. In the current package the local Biot–Savart coefficient $A_K \to 1/(4\pi)$ satisfies this standard. The pressure prefactor

TABLE II. Numerical values of all closed quantities from CODATA 2022 inputs [3]. The fourth column gives the propagated fractional uncertainty from R_∞ (1.9×10^{-12}) where applicable; dashes indicate quantities with no CODATA counterpart.

Quantity	Symbol	Value	Prop. unc. from R_∞
Circ. speed	$\lVert\mathbf{v}_\circlearrowleft\rVert$	$1.093\,845\,63 \times 10^{6}\,\mathrm{m\,s^{-1}}$	$\sim 1 \times 10^{-12}$
Core radius	r_c	$1.408\,970\,16 \times 10^{-15}\,\mathrm{m}$	$\sim 1 \times 10^{-12}$
$\tilde{\alpha}$		$7.297\,352\,567 \times 10^{-3}$	$\sim 1 \times 10^{-12}$
Circulation	Γ_0	$9.683\,619 \times 10^{-9}\,\mathrm{m^2\,s^{-1}}$	—
Line tension	$\mathcal{T}_c$	$29.053\,51\,\mathrm{N}$	—
Core density	ρ_{core}	$3.8934 \times 10^{18}\,\mathrm{kg\,m^{-3}}$	—
Geom. gate	$4/\tilde{\alpha}$	548.14	$\sim 1 \times 10^{-12}$
Spherical-cell FSC	$\alpha_{\mathrm{cell}}^{-1}$	137.035 999 211	closure-systematic
Spherical factor	Ξ_{sph}	1.046 044 51	closure-systematic

$16\pi/3$, the finite-shell coefficient $11/48$, the cell-radius closure $R_{\mathrm{cell}} = 2L_K$, and the far-field stiffness normalization $\mathcal{K}_{\mathrm{cell}} = E_{\mathrm{eff}}/(8\pi)$ remain explicit derived-label gates. Until those gates are passed, the companion result should be described as a closure-derived dimensionless fine-structure-scale eigenvalue, not as a complete first-principles derivation of QED α.

XII. DISCUSSION

The central result is a closure analysis, not a first-principles derivation. The Compton-core postulate fixes the relation between rotation rate, speed, and radius. The Rydberg anchor fixes the speed only because it is equivalent to the fine-structure constant. The line tension and density then follow only after two further mechanical closures are imposed.

This organization removes the circularity in the stronger claim that α or ρ_{core} had been derived. The paper instead identifies exactly where new dynamics would be required. The most important missing piece is an independent derivation of $\tilde{\alpha} = 2\lVert\mathbf{v}_\circlearrowleft\rVert/c$ from a specified interaction functional. The second missing piece is an on-shell core-energy calculation yielding $\mathcal{T}_c = m_e c^2/(2r_c)$. The third is a stress-balance derivation of the thin-wall Laplace law.

The spherical pressure-cell construction of Sec. VIII B shows one explicit route by which the speed closure could become predictive: the local trefoil coefficient fixes $a = r_c$, and the spherical volume-curvature response fixes the residual aspect correction. However, because the cell functional is still postulated, the result is best described as a conjectural closure theorem rather than as a derivation from the primitive field equations. Appendices C and F make explicit which part is geometrical and which part remains model-dependent.

The background fluid density ρ_f remains open. Its determination requires an independent electromagnetic–circulatory coupling or another observable normalization. It is therefore deliberately excluded from the claimed closure chain.

XIII. CONCLUSION

A Compton-core vortex-filament model becomes algebraically closed once four ingredients are supplied: the Compton-core rotation postulate, the spectroscopic speed closure, the core-diameter line-tension normalization, and the thin-wall pressure-balance closure. These give

$$\begin{aligned} \|\mathbf{v}_{\circlearrowleft}\| &= \sqrt{\pi\hbar c R_\infty/m_e}, & r_c &= \|\mathbf{v}_{\circlearrowleft}\|/\omega_C = r_e/2, \\ \Gamma_0 &= 2\pi r_c \|\mathbf{v}_{\circlearrowleft}\|, & \mathcal{T}_c &= m_e c^2/(2r_c), \\ \rho_{\text{core}} &= m_e c^2/(2\pi\|\mathbf{v}_{\circlearrowleft}\|^2 r_c^3). \end{aligned}$$

The relations are dimensionally consistent and internally coherent, but only the first is a structural postulate and the others depend on explicit closures. A future first-principles derivation must replace those closures by a dynamical calculation of α, the line tension, the interfacial stress law, and at least one low-energy observable coefficient.

The spherical pressure-cell functional added here provides one explicit candidate for the α closure, Eq. (35). It succeeds algebraically and numerically within the stated pressure-cell model, but its status remains conditional on deriving that functional from the full hydrodynamic or field-theoretic SST dynamics.

Appendix A: Assumption ledger and epistemic status

This appendix gives the audit ledger for the closure chain. The purpose is to separate algebraic consequences from empirical calibrations and from model-specific pressure-cell closures. Standard constants and the fine-structure constant are understood in their CODATA 2022 sense [3]; the electromagnetic coupling interpretation of α is the standard one used in quantum electrodynamics and particle-data summaries [21, 22]. Vortex-filament and local-induction terminology follows the classical hydrodynamic literature [4, 9, 16].

TABLE III. Assumption ledger. "Input" means externally supplied; "closure" means model-specific; "derived" means algebraic consequence within the listed assumptions.

Label	Statement	Status	Main role
A0	$m_e, c, \hbar, R_\infty, \alpha$	empirical constants	reference scales
A1	$\Omega_{\rm core} = m_e c^2/\hbar$	core postulate	Compton rotation
A2	$2\|\mathbf{v}_{\circlearrowleft}\|/c = \tilde{\alpha}$	spectroscopic closure	speed calibration
A3	$\mathcal{T}_c = m_e c^2/(2r_c)$	energy closure	line tension
A4	$P_L = \gamma_c/r_c$	thin-wall closure	density
A5	BS logarithmic self-energy	filament ansatz	local pressure
A6	$A_K \to 1/(4\pi)$	conjectured slender-core asymptote	$a = r_c$
A7	$\mathcal{E}_0 = 274.074996$	conjectured external aspect input	leading FSC scale
A8	$R_{\rm cell} = \chi_R L_K$, $\chi_R = 2$	spherical-cell closure	outer-cell scale
A9	$\eta_K = 1/(2\chi_R \mathcal{L}_K)$	algebraic from A8	shell parameter
A10	$\Xi_{\rm sph} = 1 + 3\eta_K + \eta_K^2$	pressure-cell closure	aspect correction

Under A0–A4, the paper proves only the algebraic closure of $\|\mathbf{v}_{\circlearrowleft}\|$, r_c, Γ_0, $\mathcal{T}_c$, and $\rho_{\rm core}$ from the chosen inputs. Under A5–A10, it proves the conditional spherical-cell reproduction of $\alpha_{\rm cell}$. The strongest defensible statement is therefore not that α is derived from first principles, but that the proposed cell model reproduces its numerical scale from explicit geometric, pressure-balance, and aspect inputs. Assumption A7 is numerically dominant: since $\alpha_{\rm cell} \simeq 2/\mathcal{E}_0$, it fixes the first four to five significant figures of the result. A fundamental derivation requires replacing A7 and A8 by an independent eigenvalue/stationarity problem. The companion BEM/NLS-shell finite-cell calculation is one candidate solution of that problem; in the present paper A7 and A8 remain ledgered as closure inputs, because the purpose here is to audit what the Compton-core algebra itself fixes. The numerical ideal-trefoil ropelength used here is taken from the Knot Atlas ideal-knot data file [18]; rigorous ropelength theory and bounds are discussed in Ref. [17].

Appendix B: Biot–Savart pressure-balance theorem

The local part of the self-energy of a slender filament has a logarithmic cutoff dependence. In classical vortex dynamics this is the same structural origin as the local-induction logarithm [9, 16]. Here the normalized form is written as

$$E_{\mathrm{BS}}(a) = \rho_f \Gamma_0^2 L_K \left[A_K \log\left(\frac{L_K}{a}\right) + a_K \right], \tag{B1}$$

where a is the tube radius and $2\pi a L_K$ is the tube-wall area. The normal pressure generated by this self-energy is

$$\begin{aligned} P_{\mathrm{irr}}(a) &= -\frac{1}{2\pi a L_K} \frac{\partial E_{\mathrm{BS}}}{\partial a} \\ &= \frac{\rho_f \Gamma_0^2 A_K}{2\pi a^2}. \end{aligned} \tag{B2}$$

The sign convention is that positive P_{irr} opposes an inward decrease of the tube radius.

Balancing this pressure against the core dynamic pressure,

$$P_{\mathrm{core}} = \frac{1}{2} \rho_f \|\mathbf{v}_{\circlearrowleft}\|^2, \tag{B3}$$

gives

$$\frac{\rho_f \Gamma_0^2 A_K}{2\pi a^2} = \frac{1}{2} \rho_f \|\mathbf{v}_{\circlearrowleft}\|^2. \tag{B4}$$

Using the circulation definition

$$\Gamma_0 = 2\pi r_c \|\mathbf{v}_{\circlearrowleft}\| \tag{B5}$$

yields

$$\boxed{\frac{a}{r_c} = \sqrt{4\pi A_K}}. \tag{B6}$$

Thus the conjectured slender-core asymptote $A_K = 1/(4\pi)$ is exactly the no-contact core-radius closure $a = r_c$. The units are consistent because $\rho_f \Gamma_0^2 / a^2$ has units $(\mathrm{kg\,m^{-3}})(\mathrm{m^4\,s^{-2}})\mathrm{m^{-2}} = \mathrm{Pa}$.

Appendix C: Spherical pressure-cell geometry

This appendix gives the geometrical origin of the cell factor used in Sec. VIII B. The construction is a coarse-grained control-volume model, not a theorem of classical hydrodynamics.

The spherical parallel-surface expansion is a standard result of convex and integral geometry [19]; here only the elementary spherical case is required.

Let R_{cell} be the radius of the surrounding spherical equilibration cell and let a be the tube radius. Define

$$\eta \equiv \frac{a}{R_{\mathrm{cell}}}. \tag{C1}$$

For a spherical surface shifted outward by a, the area Jacobian is

$$J_A(\eta) = \frac{4\pi(R_{\mathrm{cell}}+a)^2}{4\pi R_{\mathrm{cell}}^2} = (1+\eta)^2 = 1 + 2\eta + \eta^2. \tag{C2}$$

The corresponding first radial volume response per unit area is normalized as

$$J_V^{(r)}(\eta) = \eta. \tag{C3}$$

The proposed pressure-cell multiplier is therefore

$$\Xi_{\mathrm{sph}}(\eta) = J_A(\eta) + J_V^{(r)}(\eta) = 1 + 3\eta + \eta^2. \tag{C4}$$

The additional η term is the single isotropic compression/expansion mode of the surrounding spherical control volume; without it one would keep only the parallel-surface area response.

For the ideal trefoil, let

$$\mathcal{L}_K = \frac{L_K}{D_K} \tag{C5}$$

be the ropelength, with $D_K = 2a$ the tube diameter. Write the coarse-grained equilibration radius as

$$R_{\mathrm{cell}} = \chi_R L_K. \tag{C6}$$

Then

$$\eta_K(\chi_R) = \frac{a}{R_{\mathrm{cell}}} = \frac{D_K/2}{\chi_R L_K} = \frac{1}{2\chi_R \mathcal{L}_K}. \tag{C7}$$

The main text uses the closure choice $\chi_R = 2$, giving

$$\eta_K = \frac{1}{4\mathcal{L}_K}. \tag{C8}$$

For this choice,

$$\boxed{\Xi_{\mathrm{sph}} = 1 + \frac{3}{4\mathcal{L}_K} + \frac{1}{16\mathcal{L}_K^2}}. \tag{C9}$$

The nontrivial model choices in this appendix are the unweighted combination $J_A + J_V^{(r)}$ and the spherical coarse-graining coefficient $\chi_R = 2$; Appendix F shows how changes in these choices move the predicted coupling.

Appendix D: Conditional fine-structure closure theorem

Combining Appendix B and Appendix C gives the conditional closure used in the main text. Let the zeroth-order aspect parameter be $\mathcal{E}_0$ and let the effective aspect parameter be

$$\mathcal{E}_{\rm eff} = \mathcal{E}_0(1-\delta_{\rm cell}). \tag{D1}$$

The conjectured slender-core asymptote $A_K = 1/(4\pi)$ converts the finite-core factor $\pi^2 A_K/\mathcal{E}_0^2$ into

$$\frac{\pi^2 A_K}{\mathcal{E}_0^2} = \frac{\pi}{4\mathcal{E}_0^2}. \tag{D2}$$

Multiplication by the spherical-cell factor gives

$$\delta_{\rm cell} = \frac{\pi}{4\mathcal{E}_0^2}\left(1+\frac{3}{4\mathcal{L}_K}+\frac{1}{16\mathcal{L}_K^2}\right). \tag{D3}$$

The conditional prediction is therefore

$$\boxed{\alpha_{\rm cell} = \frac{2}{\mathcal{E}_0\left[1-\dfrac{\pi}{4\mathcal{E}_0^2}\left(1+\dfrac{3}{4\mathcal{L}_K}+\dfrac{1}{16\mathcal{L}_K^2}\right)\right]}}. \tag{D4}$$

This is a dimensionless statement conditional on the external aspect input $\mathcal{E}_0$ and the cell-radius closure $\chi_R = 2$. It should be compared with the CODATA recommended value of α, not treated as a replacement for precision QED or atomic-recoil determinations [3, 14].

For

$$\mathcal{E}_0 = 274.074996, \qquad \mathcal{L}_K = 16.371498, \qquad \chi_R = 2, \tag{D5}$$

one obtains

$$\begin{aligned}
\eta_K &= 0.0152704413,\\
\Xi_{\rm sph} &= 1.046044510,\\
\mathcal{E}_{\rm eff} &= 274.071998421,\\
\alpha_{\rm cell}^{-1} &= 137.035999211.
\end{aligned} \tag{D6}$$

The agreement with CODATA 2022 is at the 10^{-10} fractional level, but only within the pressure-cell assumptions listed in Appendix A. The leading numerical scale is fixed by $\mathcal{E}_0$, not by the spherical correction. The ideal-trefoil ropelength value is taken from the Knot Atlas ideal-knot data file [18]; the minimum-ropelength framework is discussed in Ref. [17].

Appendix E: Numerical reproducibility and convergence requirements

The numerical role of A_K is independent of the spherical-cell ansatz. It is extracted from the local logarithmic slope of $E_{\rm BS}(a)/L_K$ against $-\log a$, as described in classical local-induction calculations [9]. The robust implementation solves stationary roots in $x = a/r_c$ rather than in meters to avoid absolute-tolerance artifacts at 10^{-15} m scales.

TABLE IV. Representative trefoil closure outputs. The first line is an early 30-mode local-plateau extraction; the second line is the higher-resolution v10.3 practical estimate using the ideal database and local C++ scan.

Run	Geometry/source	A_K	$A_K/(1/4\pi)$	$a_{\rm nc}/r_c$
30-mode plateau	fallback ideal trefoil	0.08330250	1.046810	1.0231
v10.3 practical	ideal.txt, 183 modes	0.07957157	0.99992579	0.99996290

The practical estimate is therefore consistent with

$$A_K \to \frac{1}{4\pi}, \qquad \frac{a_{\rm nc}}{r_c} = \sqrt{4\pi A_K} \to 1. \tag{E1}$$

For a publication-grade numerical theorem, the remaining requirements are: (i) monotone or contractive high-N tails for the preferred plateau method; (ii) agreement between local-slope and global-fit diagnostics; (iii) mesh and regularization independence; and (iv) a public script and data archive giving $N_{\rm geom}$, $N_{\rm int}$, A_K, $A_K/(1/4\pi)$, and $a_{\rm nc}/r_c$ for every retained run. Until those checks are satisfied, $A_K = 1/(4\pi)$ should be stated as a conjectured asymptotic value or robust numerical-limit ansatz, not as a proven analytical identity.

Appendix F: Sensitivity and falsification

The spherical-cell closure is falsifiable because its coefficients enter $\alpha_{\rm cell}$ explicitly. First replace the quadratic term by a free coefficient,

$$\Xi_{\rm sph}(\beta) = 1 + 3\eta_K + \beta\eta_K^2. \tag{F1}$$

Then the prediction becomes

$$\alpha_{\rm cell}(\beta) = \frac{2}{\mathcal{E}_0\left[1 - \frac{\pi}{4\mathcal{E}_0^2}\left(1 + 3\eta_K + \beta\eta_K^2\right)\right]}. \tag{F2}$$

For the numerical choices of Eq. (D5), the exact CODATA central value would require

$$\beta_{\text{req}} \simeq 1.10, \tag{F3}$$

whereas the spherical parallel-surface model uses $\beta = 1$. Thus the agreement is good but not insensitive to the assumed second-order cell geometry. This is a key falsifier: a different coarse-grained cell geometry predicts a different β and therefore a different α_{cell}.

Second, expose the cell-radius choice by writing

$$\eta_K(\chi_R) = \frac{1}{2\chi_R \mathcal{L}_K}. \tag{F4}$$

The three-parameter sensitivity map is then

$$\alpha_{\text{cell}}(\mathcal{E}_0, \beta, \chi_R) = \frac{2}{\mathcal{E}_0 \left[1 - \frac{\pi}{4\mathcal{E}_0^2}\left(1 + 3\eta_K(\chi_R) + \beta\eta_K(\chi_R)^2\right)\right]}. \tag{F5}$$

This equation makes the parameter correlations explicit. A shift of $\mathcal{E}_0$ changes the leading value at order $\Delta\mathcal{E}_0/\mathcal{E}_0$, while β and χ_R affect only the residual finite-core correction at order $\mathcal{E}_0^{-2}$. Therefore the sub-ppm agreement cannot be interpreted as an independent prediction unless $\mathcal{E}_0$ and χ_R are obtained from a separate variational or spectral problem.

A third falsifier is the ropelength dependence. Differentiating Eq. (D4) shows that changing $\mathcal{L}_K$ changes α_{cell} at order $\mathcal{E}_0^{-3}\mathcal{L}_K^{-2}$. A non-trefoil candidate or a different ideal-knot ropelength must therefore move the prediction. Finally, any low-energy application must produce an observable correction such as a form factor or a short-range potential, for example

$$F_e(q^2) = 1 - \frac{1}{6}\langle r^2\rangle_{\text{eff}} q^2 + O(q^4), \tag{F6}$$

or

$$V(r) = -\frac{e^2}{4\pi\epsilon_0 r} + \Delta V(r; r_c), \tag{F7}$$

which can be constrained by atomic spectroscopy and QED precision tests [14, 15, 21].

Appendix G: Historical notes not used as evidence

This appendix records two historical numerical motifs that may motivate a search pattern but are not used as evidentiary steps in the closure theorem. First, Clausius reported gas-expansion coefficients near $0.003665\,\mathrm{K}^{-1}$ in the nineteenth-century mechanical theory of heat [23]. This is close to $\alpha/2$ numerically, but it is dimensional and depends on temperature scale and material regime; it is therefore not a derivation of a fundamental dimensionless coupling.

Second, non-standard non-holonomic-background models have discussed aspect ratios near 274 and associated velocity ratios [20, 24]. Those sources are not part of the orthodox derivation. In the present paper $\mathcal{E}_0$ is treated as the conjectured external aspect input A7 in Appendix A. If an orthodox version is desired, historical material from this appendix can be omitted without changing the algebraic pressure-cell theorem; only the explicit status of A7 must remain.

ACKNOWLEDGMENTS

The author acknowledges helpful discussions on the distinction between algebraic closure, empirical calibration, and first-principles derivation.

Companion-paper status

The present paper is intentionally conservative: it records the closure inputs needed by the Compton-core scaling model. Companion work supplies a proposed BEM/NLS-shell finite-cell eigenvalue calculation for the previously external aspect scale, and a separate far-field theorem target for identifying α_{cell} with the electromagnetic fine-structure coupling. Those companion results strengthen the overall program but do not alter the ledger classification of the assumptions used in this paper.

[1] H. von Helmholtz, On the integrals of the hydrodynamical equations which express vortex-motion, *Philosophical Magazine* **33**, 485 (1867).

[2] Lord Kelvin (W. Thomson), On vortex atoms, *Proceedings of the Royal Society of Edinburgh* **6**, 94 (1867).

[3] P. J. Mohr, D. B. Newell, B. N. Taylor, and E. Tiesinga, CODATA recommended values of the fundamental physical constants: 2022, arXiv:2409.03787 (2024).

[4] R. J. Donnelly, *Quantized Vortices in Helium II*, Cambridge University Press, 1991.

[5] C. J. Pethick and H. Smith, *Bose–Einstein Condensation in Dilute Gases*, 2nd ed., Cambridge University Press, 2008. doi:10.1017/CBO9780511802850

[6] L. D. Faddeev and A. J. Niemi, Stable knot-like structures in classical field theory, *Nature* **387**, 58 (1997). doi:10.1038/387058a0

[7] C. Barceló, S. Liberati, and M. Visser, Analogue gravity, *Living Reviews in Relativity* **8**, 12 (2005). doi:10.12942/lrr-2005-12

[8] G. E. Volovik, *The Universe in a Helium Droplet*, Oxford University Press, 2003.

[9] P. G. Saffman, *Vortex Dynamics*, Cambridge University Press, 1992. doi:10.1017/CBO9780511624063

[10] L. Onsager, Statistical hydrodynamics, *Nuovo Cimento Supplement* **6**, 279 (1949). doi:10.1007/BF02780991

[11] R. P. Feynman, Application of quantum mechanics to liquid helium, in *Progress in Low Temperature Physics*, vol. 1, ed. C. J. Gorter, p. 17, North-Holland, 1955.

[12] G. W. Gibbons, The maximum tension principle in general relativity, *Foundations of Physics* **32**, 1891 (2002). doi:10.1023/A:1022370717626

[13] C. Schiller, Simple derivation of minimum length, minimum dipole moment, and lack of space-time continuity, *International Journal of Theoretical Physics* **45**, 221 (2006). doi:10.1007/s10773-005-4835-2

[14] S. G. Karshenboim, Precision physics of simple atoms: QED tests, nuclear structure, and fundamental constants, *Physics Reports* **422**, 1 (2005). doi:10.1016/j.physrep.2005.08.008

[15] H. A. Bethe, The electromagnetic shift of energy levels, *Physical Review* **72**, 339 (1947). doi:10.1103/PhysRev.72.339

[16] G. K. Batchelor, *An Introduction to Fluid Dynamics*, Cambridge University Press, 1967. doi:10.1017/CBO9780511800955

[17] J. Cantarella, R. B. Kusner, and J. M. Sullivan, On the minimum ropelength of knots and links, *Inventiones Mathematicae* **150**, 257 (2002). doi:10.1007/s00222-002-0234-y

[18] D. Bar-Natan and contributors, The Knot Atlas: ideal knot data file `Ideal.txt.gz`, https://katlas.org/images/d/d2/Ideal.txt.gz. Accessed 2026.

[19] R. Schneider, *Convex Bodies: The Brunn–Minkowski Theory*, 2nd ed., Cambridge University Press, 2014. doi:10.1017/CBO9781139003858

[20] D. Rabounski and L. Borissova, A theory of the Podkletnov effect based on general relativity: anti-gravity force due to the perturbed non-holonomic background of space, *Progress in Physics* **3**, 57 (2007). https://www.ptep-online.com/2007/PP-10-07.PDF

[21] R. L. Workman et al. (Particle Data Group), Review of particle physics, *Progress of Theoretical and Experimental Physics* **2022**, 083C01 (2022). doi:10.1093/ptep/ptac097

[22] M. E. Peskin and D. V. Schroeder, *An Introduction to Quantum Field Theory*, Addison-Wesley, 1995. ISBN: 978-0201503975.

[23] R. Clausius, *The Mechanical Theory of Heat, with its Applications to the Steam-Engine and to the Physical Properties of Bodies*, edited by T. Archer Hirst, with an introduction by J. Tyndall, John Van Voorst, London, 1867.

[24] R. Oros di Bartini, Relations between physical constants, *Progress in Physics* **3**, 34 (2005). https://www.ptep-online.com/2005/PP-03-06.PDF